Do not write outside the box

0 4 This question is about elements and compounds.

0 4 . 1 Magnesium and oxygen react to produce magnesium oxide.

Balance the equation for the reaction.

[1 mark]

$$_____\,Mg + O_2 \rightarrow 2\,MgO$$

0 4 . 2 Suggest **one** safety precaution that should be taken when heating magnesium and oxygen.

[1 mark]

_____

_____

0 4 . 3 Calculate the relative formula mass (M_r) of magnesium fluoride (MgF_2).

Relative atomic masses (A_r): F = 19 Mg = 24

[2 marks]

_____

_____

_____

_____

Relative formula mass (M_r) = _____

0 4 . 4 Argon is a noble gas.

Explain why **no** product is formed when magnesium and argon are heated together.

[2 marks]

_____

_____

_____

_____

Do not write outside the box

0 4 . 5 **Figure 10** shows a reactivity series.

Figure 10

Most reactive

Metal **D**

Sodium

Magnesium

Carbon

Metal **E**

Iron

Hydrogen

Copper

Least reactive

Do not write outside the box

Draw **one** line from each metal to the method used to extract that metal.

Use **Figure 10**.

[2 marks]

Metal	**Method used to extract that metal**
	Extracted by electrolysis of a molten ionic compound.
Metal **D**	Extracted from its oxide by reduction with carbon.
	Extracted from its oxide by reduction with hydrogen.
Metal **E**	
	Removed from the Earth as the metal itself.

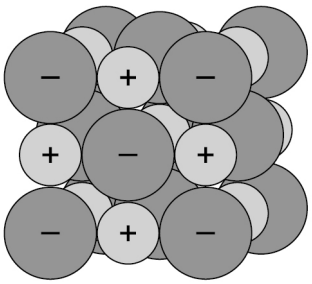
Question 4 continues on the next page

Do not write outside the box

A substance conducts electricity when it has charged particles that are free to move.

0 4 . 6 **Figure 11** represents the structure of sodium chloride.

Figure 11



Sodium chloride

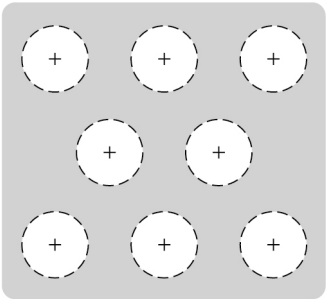
Explain why sodium chloride conducts electricity when molten but **not** when solid.

[3 marks]

Do not write outside the box

0 4 . 7 **Figure 12** represents the structure of sodium metal.

Figure 12



Sodium metal

Explain why sodium metal conducts electricity when solid.

[2 marks]

_____

_____

_____

_____

13

Turn over for the next question